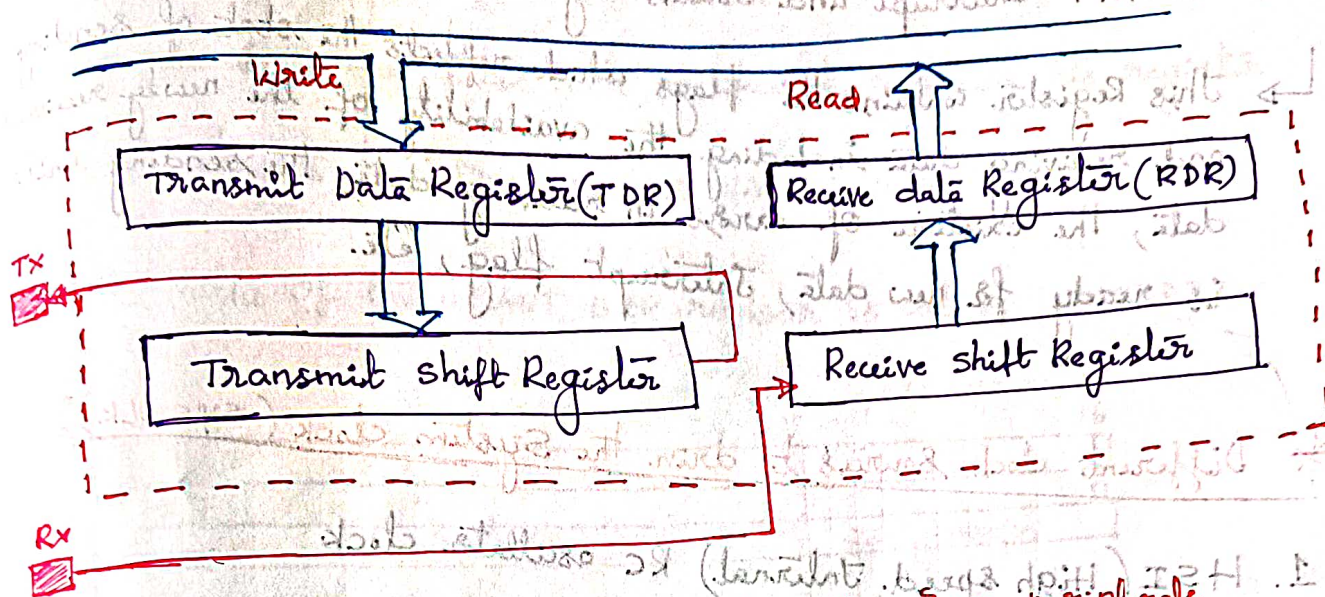


* Simplified Block diagram of USART:



→ There are three groups of Registers in USART peripherals

1) Configuration Registers:

- Before using the USART peripheral the configuration Registers must be initialized.
- Configuration parameters are Baud rate, word length, stop bit and Interrupts (if needed).
- In STM ARM microcontroller, some of the configuration Registers are BAUD, control 1 and control 2.

2) Transmit and Receive Register:

- To send data, we simply write to the Transmit Data register (USART_TDR).
- The content of Transmit data Register (TDR) is sent out through the serial transmit pin (TXD).
- The received data is stored into the Receive data register (USART_RDR) through the serial Receive pin (RXD).
- Both TDR and RDR have two different addresses.
- Write to the transmit register and read from the Receive register.

3) Interrupt and status Register: (USART_ISR)

↳ USART Interrupt and status Register

↳ This Register contains the flags which reflects the state of sending and receiving data including the availability of the newly received data, the existence of errors in receiving data, the sending unit is ready for new data, Interrupt flag, etc.

* Different clock sources to drive the System clock (SYSCLK):

1. HSI (High speed Internal) RC oscillator clock

2. HSE (High speed External) oscillator clock using a crystal
Value of 4 MHz to 26 MHz

3. PLL clock.

* The SYSCLK (system clock) is the clock that runs the peripherals such as USART. The SYSCLK itself can come from any of the above sources.

↳ The HSI clock signal is generated from an Internal 16 MHz RC oscillator and can be used directly as a system clock or used as a PLL input.

* The HSI oscillator is the default clock source for SYSCLK and is turned on upon reset.